

GRISSOM AFB, IN, USA

WMO: 725335

Lat: 40.650N

Lon: 86.150W

Elev: 248

StdP: 98.39

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94833

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-16.1	-23.0	0.5	-18.4	-19.5	0.7	-14.5	12.9	-3.0	12.0	-4.8	4.0	280	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.4	24.0	31.1	23.4	29.8	22.7	26.2	29.7	25.2	28.8	24.3	27.6	5.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	21.2	27.7	24.2	19.7	27.0	23.4	18.7	26.1	82.6	29.7	78.5	28.8	74.7	27.6	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	10.3	8.8	DB	-23.1	34.4	4.0	1.9	-26.0	35.7	-28.4	36.8	-30.6	37.8	-33.6	39.2
			WB	-23.3	27.2	4.0	1.1	-26.2	28.0	-28.6	28.6	-30.9	29.2	-33.8	30.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-4.7	-2.0	4.3	10.8	17.3	21.9	23.3	22.3	19.0	12.2	5.0	-0.4
	DBStd	10.96	7.12	6.69	6.16	5.04	4.90	3.23	3.18	2.67	4.04	5.02	5.46	5.88
	HDD10.0	1572	458	338	197	49	3	0	0	0	0	34	166	326
	HDD18.3	3264	716	568	437	230	81	8	3	3	40	199	399	581
	CDD10.0	1869	1	3	21	74	229	357	411	381	270	103	17	3
	CDD18.3	519	0	0	1	4	48	115	156	126	59	9	0	0
	CDH23.3	4318	0	0	12	45	423	896	1389	978	498	77	1	0
	CDH26.7	1342	0	0	2	3	118	275	492	277	162	14	0	0
Wind	WSAvg	4.2	5.0	4.9	4.7	5.0	4.2	3.7	3.0	2.8	3.2	4.3	4.5	4.7
Precipitation	PrecAvg	1112	67	58	75	104	113	131	123	106	97	81	83	74
	PrecMax	1528	177	146	149	254	196	372	355	201	209	179	167	170
	PrecMin	766	22	11	18	27	35	29	38	26	23	25	20	21
	PrecStd	184	32	35	31	51	40	66	65	42	44	44	33	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	17.8	26.1	27.2	32.2	33.4	35.4	32.4	33.1	28.7	21.9	16.8
		MCWB	12.9	13.3	16.5	17.5	21.8	23.0	25.9	24.2	23.1	20.2	14.2	14.7
	2%	DB	10.8	14.2	20.2	24.5	29.6	31.5	32.7	30.9	31.0	25.6	18.7	13.9
		MCWB	10.0	12.8	15.1	16.6	20.9	23.3	25.2	23.8	22.9	18.4	14.2	13.0
	5%	DB	7.6	10.8	16.8	22.2	27.8	30.0	30.9	29.7	28.7	22.8	15.8	10.4
		MCWB	6.6	9.6	13.4	15.6	20.1	22.5	23.9	23.2	21.5	17.6	13.1	9.4
	10%	DB	4.2	7.2	13.9	19.6	25.8	28.1	29.4	28.2	26.3	20.6	13.0	7.4
		MCWB	3.4	6.0	11.0	14.0	19.5	21.5	23.3	22.5	20.0	16.5	10.0	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.7	18.0	19.9	24.6	26.4	27.9	26.8	25.2	23.0	17.6	15.5
		MCDB	13.8	16.6	20.1	23.5	27.4	30.3	31.1	29.6	29.7	26.0	20.1	16.5
	2%	WB	10.2	12.5	16.3	18.3	22.8	25.2	26.6	25.6	24.1	21.0	15.5	12.7
		MCDB	10.6	13.8	20.5	21.9	26.2	28.9	30.1	28.8	28.4	22.8	17.0	13.2
	5%	WB	6.5	9.5	14.1	16.9	21.7	24.1	25.6	24.6	23.1	19.1	12.9	9.5
		MCDB	7.3	10.8	16.2	20.2	25.5	27.1	29.3	27.7	26.7	21.4	15.7	10.2
	10%	WB	3.5	5.8	11.4	15.3	20.6	23.1	24.7	23.7	21.9	17.1	10.7	6.6
		MCDB	4.3	7.1	13.2	18.7	24.3	26.4	28.1	26.5	24.5	19.7	12.9	7.1
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	10.0	11.4	11.0	10.4	10.5	10.7	11.3	10.6	9.1	7.3
		MCDBR	10.3	12.3	13.4	14.4	12.9	12.0	11.7	11.7	13.3	13.4	12.8	10.5
		MCWBR	9.3	10.3	9.2	8.5	6.7	5.7	5.1	5.5	5.6	7.0	8.5	9.3
	5% WB	MCDBR	10.1	11.8	11.8	12.3	11.0	9.9	10.3	9.8	10.6	10.7	11.5	10.0
		MCWBR	9.6	10.4	8.9	9.0	6.9	5.7	5.2	5.4	5.3	7.7	8.9	9.4
Clear-Sky Solar Irradiance	taub		0.299	0.326	0.347	0.410	0.449	0.437	0.452	0.453	0.416	0.358	0.320	0.301
	taud		2.410	2.323	2.359	2.187	2.124	2.208	2.173	2.156	2.240	2.400	2.455	2.439
	Ebn at Noon		860	882	903	862	833	842	825	812	818	837	828	827
	Edh at Noon		81	102	111	141	154	142	146	144	124	94	78	73
All-Sky Solar Radiation	RadAvg		1.67	2.53	3.61	4.69	5.38	6.11	6.11	5.43	4.46	3.03	1.97	1.38
	RadStd		0.17	0.27	0.28	0.43	0.49	0.40	0.35	0.31	0.35	0.28	0.23	0.15

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page